Permutation Rounds

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3398 | Mathematics | https://cses.fi/problemset/task/3398

There is a sorted array [1,2,…,n] and a permutation $p_1,p_2,\ldots,p_n$. On each round, all elements move according to the permutation: the element at position i moves to position p_i.

After how many rounds is the array sorted again for the first time?

Input

The first line has an integer n.

The next line contains n integers $p_1,p_2,\ldots,p_n$.

Output

Print the number of rounds modulo 10^9+7.

Constraints

- $1 \le n \le 2 \cdot 10^5$

Example

Input
8 5 3 2 6 4 1 8 7

Output
4

Explanation: The array changes as follows after the rounds:

- Round 1: [6,3,2,5,1,4,8,7]
- Round 2: [4,2,3,1,6,5,7,8]
- Round 3: [5,3,2,6,4,1,8,7]
- Round 4: [1,2,3,4,5,6,7,8]